

Kinetic versus thermodynamic products

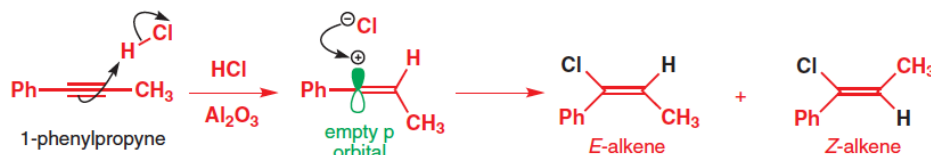
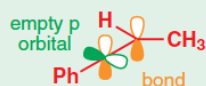
We started this chapter with a discussion of thermodynamics: the factors that govern equilibria. We then moved onto rates: the factors that determine the *rate* at which reactions proceed. Depending on the reaction, either may be more important, and in general:

- Reactions under **thermodynamic control** have outcomes that depend on the **position of an equilibrium** and therefore the **relative stability of the possible products**.
- Reactions under **kinetic control** have outcomes that depend on the **rate at which the reaction proceeds**, and therefore on the **relative energies of the transition states leading to the alternative products**.

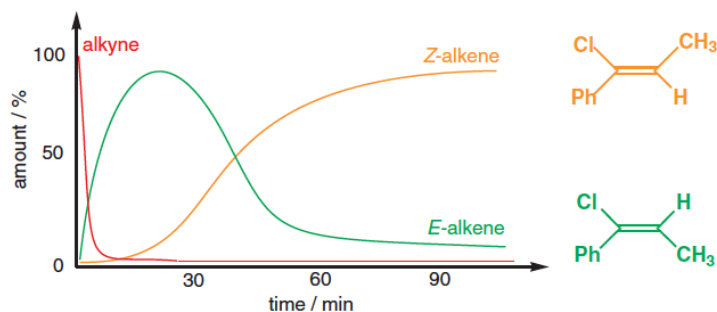
➡ There are further examples of contrasting kinetic and thermodynamic control in Chapters 19 and 22.

■ Hydrogen chloride is a gas, but it can be absorbed onto the surface of the alumina for convenient handling.

■ It's worth taking a moment to think about the structure of the intermediate cation here: the cationic carbon is sp hybridized (linear) with an empty p orbital perpendicular to the p orbitals of the double bond (it is the p orbital that used to be involved in the second π bond of the alkyne).



The two alkenes are labelled *E* and *Z*. After about 2 hours the main product is the *Z*-alkene. However, this is not the case in the early stages of the reaction. The graph below shows how the proportions of the starting material and the two products change with time.



Points to note:

- When the alkyne concentration drops almost to zero (10 minutes), the only alkene that has been formed is the *E*-alkene.

- As time increases, the amount of *E*-alkene decreases as the amount of *Z*-alkene increases.
- Eventually, the proportions of *E*- and *Z*-alkene do not change.

Since it is the *Z*-alkene that dominates at equilibrium, this must be lower in energy than the *E*-alkene. Since we know the ratio of the products at equilibrium, we can work out the difference in energy between the two isomers:

ratio of *E*:*Z*-alkenes at equilibrium = 1:35

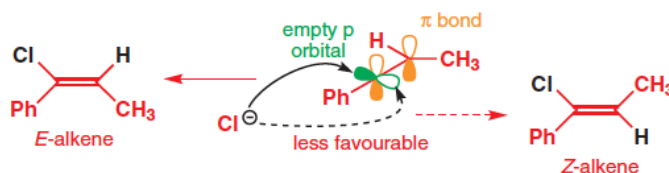
$$K_{\text{eq}} = \frac{[Z]}{[E]} = 35$$

$$\Delta G = -RT \ln K = -8.314 \times 298 \times \ln(35) = -8.8 \text{ kJ mol}^{-1}$$

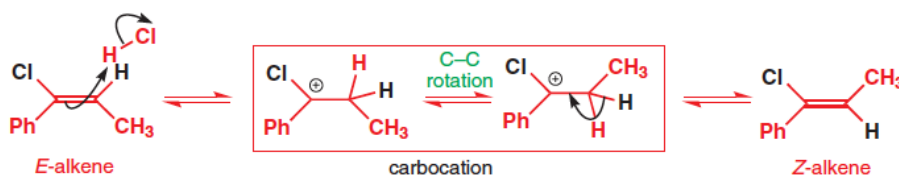
that is, the *Z*-alkene is 8.8 kJ mol⁻¹ lower in energy than the *E*-alkene.

However, although the *Z*-alkene is more stable, the *E*-alkene is formed faster under these conditions: the route to the *E*-alkene must have a smaller activation energy barrier than *trans* addition. This is quite easy to understand: the intermediate cation has no double-bond geometry because the cationic C is sp hybridized (linear). When chloride attacks, it prefers to attack from the side of the H atom rather than the (bigger) methyl group.

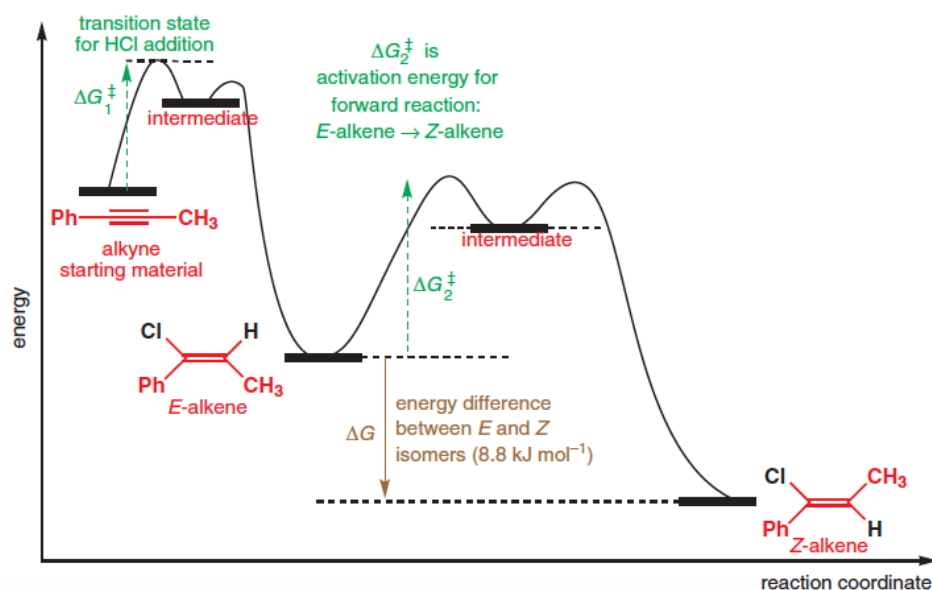
■ You might normally expect an *E*-alkene to be more stable than a *Z*-alkene—it just so happens here that Cl has a higher priority than Ph and the *Z*-alkene has the two largest groups (Ph and Me) *trans* (see p. 392 for rules of nomenclature).



There must then be some mechanism by which the quickly formed *E*-alkene is converted into the more stable *Z*-alkene. The conditions are acidic, so the most likely mechanism is the acid-catalysed alkene isomerization you saw earlier in the chapter:



This information can be summarized on an energy profile diagram:



Initially, the alkyne is converted into the *E*-alkene via the intermediate linear cation. The activation energy for this step is labelled $\Delta G_1^\ddagger$. The *E*-alkene can convert to the *Z* isomer via an intermediate, with activation energy $\Delta G_2^\ddagger$. Since $\Delta G_1^\ddagger$ is smaller than $\Delta G_2^\ddagger$, the *E*-alkene forms faster than it isomerizes, and all the alkyne is rapidly converted to the *E*-alkene. But over the course of the reaction, the *E*-alkene slowly isomerizes to the *Z*-alkene. An equilibrium is eventually reached that favours the *Z*-alkene because it is more stable (by 8.8 kJ mol^{-1} , as we calculated earlier). Why doesn't the *Z*-alkene form faster than the *E*? Well, as we suggested above, the transition state for its formation from the linear cation must be higher in energy than the transition state for formation of the *E*-alkene, because of steric hindrance.

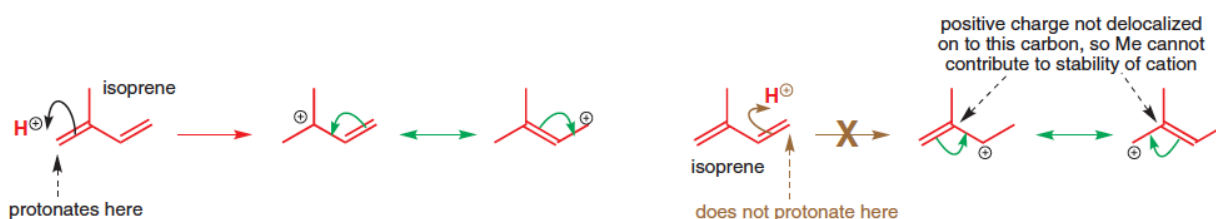
● Kinetic and thermodynamic products

- The *E*-alkene is formed faster and is known as the **kinetic product** or the **product of kinetic control**.
- The *Z*-alkene is more stable and is known as the **thermodynamic product** or the **product of thermodynamic control**.

If we wanted to isolate the kinetic product, the *E*-alkene, we would carry out the reaction at low temperature and not leave it long enough for equilibration. If, on the other hand, we want the thermodynamic product, the *Z*-alkene, we would leave the reaction for longer at higher temperatures to make sure that the larger energy barrier yielding the most stable product can be overcome.

Electrophilic addition to dienes

Earlier in the chapter you saw the epoxidation of a diene to give a monoepoxide: only one of the double bonds reacted. This is quite a usual observation: dienes are more nucleophilic than isolated alkenes. This is easy to explain by looking at the relative energy of the HOMO of an alkene and a diene—this discussion is on p. 138 of Chapter 7. Dienes are therefore very susceptible to protonation by acid to give a cation. This is what happens when 2-methylbuta-1,3-diene (isoprene) is treated with acid. Protonation gives a stable delocalized allylic cation.

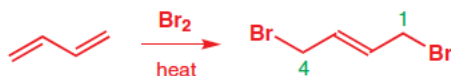


Why protonate this double bond and not the other one? The cation you get by protonating the other double bond is also allylic, but it cannot benefit from the additional stabilization from the methyl group because the positive charge is not delocalized on to the carbon carrying the methyl.

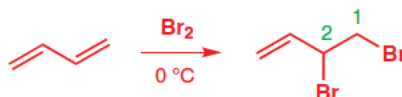
If the acid is HBr, then nucleophilic attack by Br on the cation follows. The cation is attacked at the less hindered end to give the important compound prenyl bromide. This is very much the sort of reaction you met in Chapter 15—it is the second half of an $\text{S}_{\text{N}}1$ substitution reaction on an allylic compound.



Overall, the atoms H and Br are added to the ends of the diene system. The same appears to be the case when dienes are brominated with Br_2 .



Changing the conditions slightly gives a different outcome. If the reaction is done at lower temperatures, the bromine just adds across one of the double bonds to give a 1,2-dibromide.



This compound turns out to be the kinetic product of the bromination reaction. The 1,4-dibromide is formed only when the reaction is heated, and is the thermodynamic product. The mechanism is electrophilic attack on the diene to give a bromonium ion, which bromide opens to give the dibromide. We have shown the bromide attacking the more substituted end of the bromonium ion—although we can't know this for sure (attack at either end gives the same product), you are about to see (in the next section) evidence that this is the usual course of reactions of unsymmetrical bromonium ions.

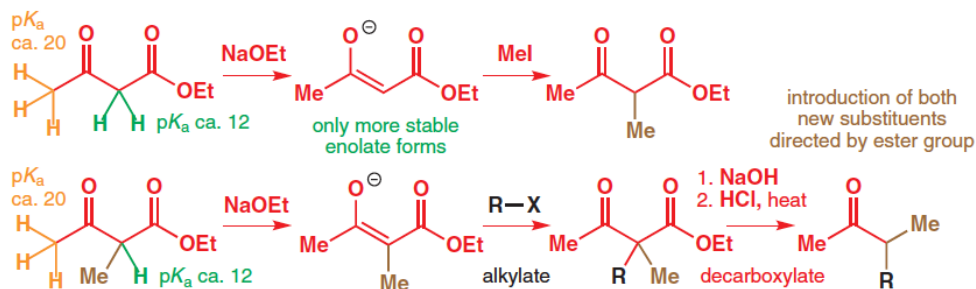


This 1,2-dibromide can still react further because it can undergo nucleophilic substitution. Bromide is a good nucleophile and a good leaving group and, with an allylic system like this, an $\text{S}_{\text{N}}1$ reaction can take place in which both the nucleophile and the leaving group are bromide. The intermediate is a cation, but here the carbocation is disguised as the bromonium ion because bromine's lone pair can help stabilize the positive charge. Bromide can attack where it left, returning to starting material, but it can also attack the far end of the allylic system, giving the 1,4-dibromide. The steps are all reversible at higher temperatures, so the fact that the 1,4-dibromide is formed under these conditions must mean it is more stable than the 1,2-dibromide. It is not hard to see why: it has a more substituted double bond and the two large bromine atoms are further apart.

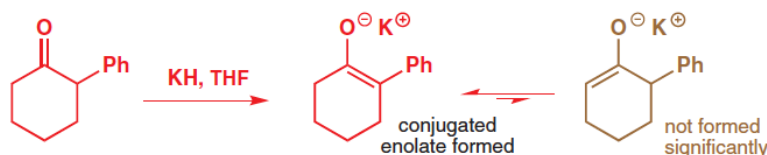


Thermodynamically controlled enolate formation

Selective enolate formation is straightforward if the protons on one side of the ketone are significantly more acidic than those on the other. This is what you have just seen with ethyl acetoacetate: it is a ketone, but with weak bases (pK_a of the conjugate acid < 18) it only ever enolizes on the side where the protons are acidified by the second electron-withdrawing group. If two new substituents are introduced, in the manner you have just seen, they will always both be joined to the same carbon atom. This is an example of thermodynamic control: only the more stable of the two possible enolates is formed.



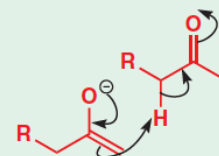
This principle can be extended to ketones whose enolates have less dramatic differences in stability. Since enols and enolates are alkenes, the more substituents they carry the more stable they are. So, in principle, even additional alkyl groups can control enolate formation under thermodynamic control. Formation of the more stable enolate requires a mechanism for equilibration between the two enolates, and this must be proton transfer. If a proton source is available—and this can even be just excess ketone—an equilibrium mixture of the two enolates will form. The composition of this equilibrium mixture depends very much on the ketone but, with 2-phenylcyclohexanone, conjugation ensures that only one enolate forms. The base is potassium hydride: it's strong, but small (and so has no difficulty removing the more hindered proton) and can be used under conditions that permit enolate equilibration.



The more substituted lithium enolates can also be formed from the more substituted silyl enol ethers by substitution at silicon—a reaction you met in Chapter 20. The value of this reaction now becomes clear because the usual way of making silyl enol ethers (Me_3SiCl , Et_3N) typically produces, from unsymmetrical ketones, the more substituted of the two possible ethers. Because the silyl enol ether (unlike the enolate itself) can be purified, fully regiochemically pure enolates can be formed in this way.

➡ The influence of substituents on the stability of alkenes was discussed on p. 394. The fact that more substituted enols are more stable was discussed in Chapter 20, p. 465.

■ The mechanism for equilibration simply involves deprotonation of a molecule of ketone (the proton source) by a molecule of enolate:



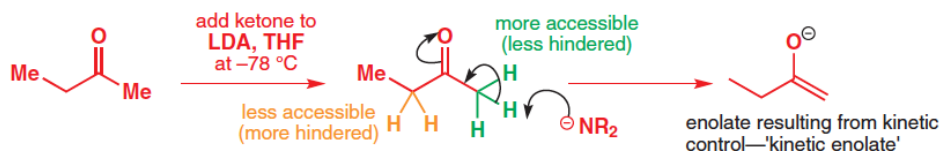
➡ Kinetic and thermodynamic control were discussed in Chapters 12, 23, and 24, pp. 264, 546, and 581.

■ To understand why less substituted C atoms have more acidic C–H bonds, think of base strengths: MeLi is a weaker base than *t*-BuLi, so the conjugate acid must be a stronger acid.

There must never be more ketone in the mixture than base, or exchange of protons between ketone and enolate will lead to equilibration. Kinetic enolate formations with LDA must be done by adding the ketone to the LDA so that there is excess LDA present throughout the reaction.

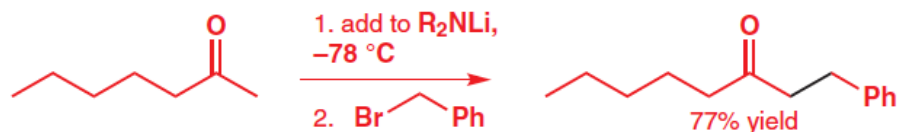
Kinetically controlled enolate formation

LDA is too hindered to attack most carbonyl C=O bonds, so it attacks C–H instead. And, if there is a choice of C–H bonds, it will attack the least hindered possible. It will also prefer to attack more acidic C–H bonds, and C–H bonds on less substituted carbons are indeed more acidic. Furthermore, statistics helps, since a less substituted C atom has more protons to be removed (three versus two in this example) so, even if the rates were the same, the less substituted enolate would predominate.

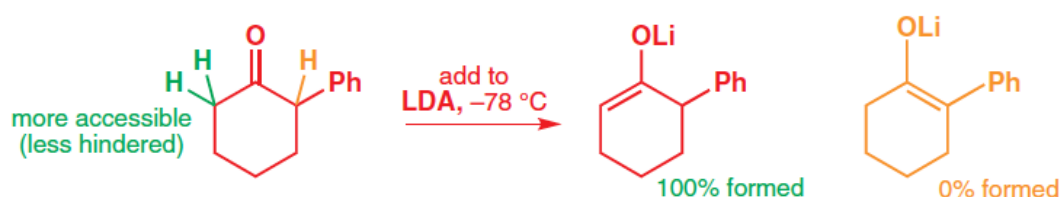


These factors multiply to ensure that the enolate that forms will be the one with the fewer substituents—provided we now prevent equilibration of the enolate to the more stable, more substituted one. This means keeping the temperature low, typically -78°C , keeping the reaction time short, and using an excess of strong base to deprotonate irreversibly and ensure that there is no remaining ketone to act as a proton source. The enolate that we then get is the one that formed faster, under kinetic control—known as the ‘kinetic enolate’—and not necessarily the one that is more stable.

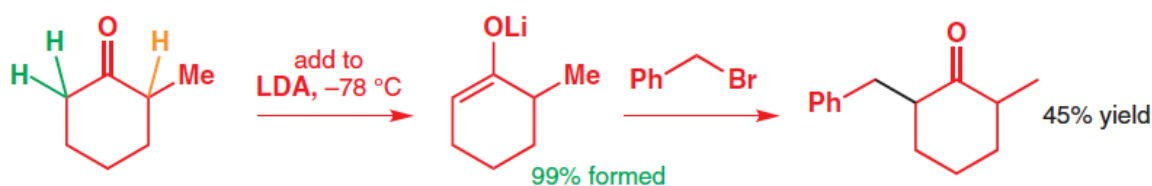
In general, this effect is sufficient to allow selective kinetic deprotonation of methyl ketones, that is, where the distinction is between Me and alkyl:



The same method works very well for 2-substituted cyclohexanones: the less substituted enolate forms. Even with 2-phenylcyclohexanone, which, as you have just seen, has a strong thermodynamic preference for the conjugated enolate, only the less substituted enolate forms.



2-Methylcyclohexanone can be regioselectively alkylated using LDA and benzyl bromide by this method.



● Regioselective formation of enolates from ketones

Thermodynamic enolates are:

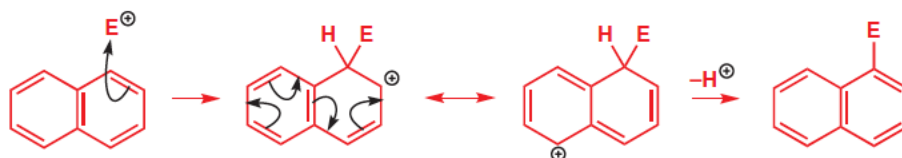
- more substituted
- more stable
- favoured by excess ketone, high temperature, long reaction time

Kinetic enolates are:

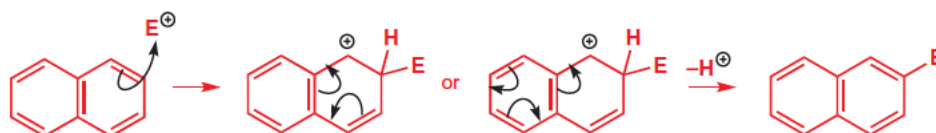
- less substituted
- less stable
- favoured by strong, hindered base (e.g. LDA), low temperature, short reaction time

Sulphonation of naphthalene

Electrophilic substitution on naphthalene normally occurs at a site next to the ring junction (α). This is because the HOMO has its largest coefficient at this atom, but you can rationalize the result by looking at the long, linear delocalization in the resulting cation, which can be represented by a single train of arrows. This extended conjugation makes naphthalene more nucleophilic than benzene. So, bromination occurs at the α -position in good yield even without a Lewis acid.



Reaction at the other position (β) is less favourable as the intermediate cation is cross-conjugated. The cation delocalizes into both rings, but no long linear chain of arrows is possible.



If the reaction is irreversible, the α -product is usually formed. But if the reaction is reversible, as is the case with sulfonation, the position of substitution may be determined by temperature. Sulfonation at low temperatures gives the α -product by kinetic control, while sulfonation at high temperatures gives the β -product by thermodynamic control. The β -product is formed more slowly but it is more stable as there is less steric hindrance between the large sulfonic acid group and the ortho hydrogen on the other ring. Under conditions allowing reversible sulfonation, eventually all the product ends up β .

